CORE 200 Scientific Methods
Handout 26

Group name

Enter your details : ______ ___

Section

Instructions/information: Following are some exercises that relate to a reading you have been assigned. Please refer to your notes as you answer them. You are encouraged to confer with your group mates as you put down your thoughts.

Module 6 | Sampling (2 of 2)

Reflect and discuss (A)

An insurance industry research foundation wants to study the quality of care given to all patients at risk for heart disease in the UnitedStates. Since not all those at risk seek treatment, the foundation randomly selects 3,500 claims only from among those at-risk patients who were actually treated for chest pain. The foundation obtains this sample in several stages. First, the foundation identifies 5 large companies that represent a broad cross-section of patients, chooses 2 of the 5 at random, and gains access to the claims of all the companies' patients. The two companies' claims are classified (depending on their origin) to 7 geographical regions (California, Florida, Great Lakes, Midwest, Northeast, Southern, and Southwest), and within each region, 5 counties are selected that represent a continuum spanning rural, suburban, and urban populations. (In total, then, patients from 35 counties are included).

Within each county (and for each company), claims of 25 male and 25 female patients treated for chest pain are randomly selected for the study, for a total of 3,500 patients.

In this study:

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1. The __________ is:

A. a diverse group of U.S. patients treated for chest pain.
B. all U.S. patients at risk for heart disease.
C. a subset of 3,500 from a diverse group of U.S. patients treated for chest pain.

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2. The __________ frame is:

A. a diverse group of U.S. patients treated for chest pain.
B. all U.S. patients at risk for heart disease.
C. a subset of 3,500 from a diverse group of U.S. patients treated for chest pain.

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3. The __________ is:

A. a diverse group of U.S. patients treated for chest pain.
B. all U.S. patients at risk for heart disease.
C. a subset of 3,500 from a diverse group of U.S. patients treated for chest pain.

For each stage below in the multistage sampling plan of this study, identify the sampling technique that was used.

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4. The research foundation identifies 5 large companies that represent a broad cross-section of patients, ______ ______ at random, and gains access to the claims of all the companies' patients.

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A. Cluster sampling.
B. Simple random sampling.
C. Stratified sampling.

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5. The __________ claims are classified (depending on their origin) according to 7 geographical regions, and within each region, the sampling continues.

A. Cluster sampling.
B. Simple random sampling.
C. Stratified sampling.

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6. From each region, 5 representative counties are selected. (In total, all the claims originating from 35 counties are examined.)

A. Cluster sampling.
B. Simple random sampling.
C. Stratified sampling.

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7. Within each county (and for each company), claims of __________ male and __________ female patients are randomly selected.

A. Cluster sampling.
B. Simple random sampling.
C. Stratified sampling.

Module 7 | Causation and Experiments (2 of 3)

Reflect and discuss (B)

8. Students in a large statistics class were randomly divided into two groups. The first group took the midterm exam with soft music playing in the background, while the second group took the exam with no music playing. The scores of the two groups on the exam were compared. This experiment is __________ because:

A. students were allowed to keep their eyes open while taking the exam.
B. the exam was too long.
C. the students know whether or not music was playing while they were taking the exam.
D. some of the students did not study for the exam.
E. students were randomized into the two groups.

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9. In an experiment to see if aspirin reduces the chance of having a heart attack, a __________ is:

A. the place where the subjects go when they have a heart attack.
B. probably administered to the control group.
C. a procedure for deciding who is going to get the aspirin treatment.
D. the randomization procedure.

Module 7 | Experiments With More Than One Explanatory Variable

Reflect and discuss (C)

A university was interested in examining the overall effectiveness of its online statistics course, along with the effectiveness of particular aspects of the course. First, the university wanted to see whether the online course was better than a standard course. Second, the university wanted to know whether students learned best using Excel, using Minitab, or using no statistical package at all. The university randomly selected a group of 30 students and administered one of the different variants of the course (i.e., traditional or online, coupled with one of the software options) to each student. The success of each variant was measured by the students' average improvement between a pre-test and a post-test.

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10. This is a:

A. controlled experiment.
B. observational study.

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11. The __________ variables are:

A. The students' mean pre-test average score.
B. Whether the course was online or traditional.
C. Whether the course was online or traditional and the statistical software used.

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12. The __________ variable is:

A. the students' mean post-test score.
B. the difference between the students' mean pre-test and post-test scores.
C. an evaluation of course quality provided by an independent rater.

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13. How many __________ are there in this study?

A. 3 B. 4 C. 5 D. 6 E. 7

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14. If we would like the treatments to be __________ in terms of the number of subjects, how many students should we assign to each treatment?

A. 3 B. 4 C. 5 D. 6 E. 7

Module 7 | Modifications to Randomization

Reflect and discuss (D)

15. Researchers wanted to study whether or not Botox injected under the arms can reduce sweating.

Which of these is a __________ design?

A. Randomly assign the underarms of some subjects to be injected with Botox, and those of other subjects with salt water.
B. For each subject, one underarm is injected with Botox, and the other with salt water.

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16. Suppose each subject has one underarm injected with Botox, and the other with salt water. Which is the best way to make the ________?

A. Inject Botox under the left arm for some participants and the right arm for others, choosing the arm randomly.
B. For the sake of consistency, inject all the right underarms with salt water and the left underarms with Botox.

Module 7 | Sample Surveys (2 of 2)

Reflect and discuss (E)

17. A midterm evaluation of a recitation instructor's teaching asked students to rate their instructor's preparation for class by choosing from one of the following: a) excellent; b) good; and, c) needs improvement. We may expect a summary of the instructor's teaching, based on the survey results, to be:

A. unbiased.
B. biased towards an unfavorable summary.
C. biased towards a favorable summary.

18. For which of the following questions would we expect the most ________ responses?

A. How tall are you?
B. How many times in your life have you used illegal drugs?
C. What is the lowest course grade you received last year?

19. What additional option should be included in the following survey question: "How often do you dream in black and white?" Options are "________,", "sometimes" and "never."

A. rarely.
B. not sure.
C. three times a week.

20. Consider this survey question: "Every year, thousands of people, including children, are ________ because of the widespread availability of firearms in the United States. Do you agree that our country needs stricter gun control laws?" The response that surveyors hope to elicit is:

A. yes.
B. no.
C. unknown.

Reflect and discuss (F)

21. The SAT-Verbal scores of a sample of 300 students at a particular university had a mean of 592 and standard deviation of 73.

According to the university's reports, the SAT-Verbal scores of all its students had a mean of 580 and a standard deviation of 110.

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22. Which of the following is a ____________?

A. 300.
B. 592.
C. 580.
D. 110.

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23. Which of the following is a ____________?

A. 300.
B. 592.
C. 110.
D. 73.

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24. ____________ are:

A. 73, 110, 592, 580.
B. 592, 580, 73, 110.
C. 580, 592, 110, 73.
D. 300, 592, 73, 110.

Reflect and discuss (G)

25. Compared to small samples, large samples have { less, more, about the same } ____________.

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26. We collect random samples of 25 students at a time and calculate the proportion of females in each sample. The standard deviation of $\hat{p}$ is approximately 0.10. Which of the following is a plausible standard deviation for samples of ____________?

A. 0.40.
B. 0.10.
C. 0.05.

Reflect and discuss (H)

27. According to the National Postsecondary Student Aid Study conducted by the U.S. Department of Education in 2008, 62% of graduates from public universities had student loans.

We randomly sample college graduates from public universities and determine the proportion in the sample with student loans. For which of the following sample sizes is a ____________ a good fit for the sampling distribution of sample proportions?

A. 10.
B. 20.
C. 30.
D. Both 20 and 30.
E. None of these.

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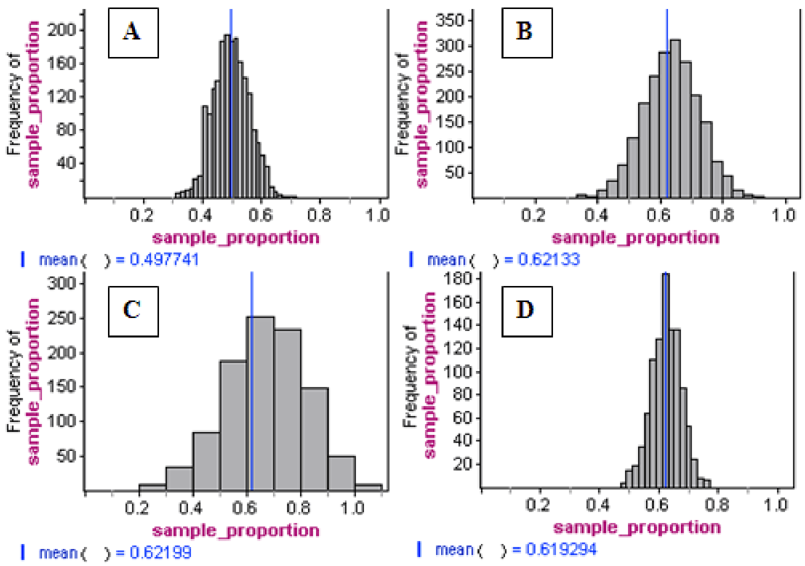
28. Which distribution is a plausible representation of the sampling distribution for random samples of ______ ______?



Reflect and discuss (I)

29. According to the official M&M website, 24% of the plain milk chocolate M&M's produced by Mars Corporation are blue. Annie buys a large family-size bag of M&M's. Sarah buys a small fun-size bag. Which bag is more likely to have more than ____________ M&M's?

A. Annie, because there are more M&M's in her bag, so she will have more blue ones.
B. Annie, because there is more variability in the proportion of blues among larger samples.
C. Sarah, because there is more variability in the proportion of blues among smaller samples.
D. Both have the same chance because they are both random samples.

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30. Imagine that you have a very large barrel that contains tens of thousands of M&M's. According to the official M&M website, 20% of the M&M's produced by the Mars Corporation are orange. 5 students each take

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a random sample of ________ M&M's and record the percentage of orange in each sample. Which sequence is the most plausible for the percentage of orange candies obtained in these 5 samples?

A. 20%, 20%, 20%, 20%, 20%.
B. 15%, 25%, 22%, 20%, 28% .
C. 5%, 80%, 8%, 65%, 70%.
D. Any of the above.

Module 8 | Behavior of Sample Proportion (3 of 3)

Reflect and discuss (J)

31. The proportion of left-handed people in the general population is about 0.1. Suppose a random sample of 225 people is observed.

What is the sampling distribution of the sample proportion ($\hat{p}$)? In other words, what can we say about the behavior of the different possible values of the sample proportion that we can get when we take such a sample?

(Note: normal approximation is ________ because $0.1(225) = 22.5$ and $0.9(225) = 202.5$ are both more than 10.)

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32. What ________ is almost certain (probability 0.997) to contain the sample proportion of left-handed people?

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33. In a sample of ________ people, would it be unusual to find that 40 people in the sample are left-handed?

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34. Find the approximate probability of at least 27 in 225 (proportion 0.12) being left-handed. In other words, what is $P(\hat{p} \geq 0.12)$?

Guidance: Note that 0.12 is exactly 1 standard deviation (0.02) above the mean (0.1). Now use the Standard Deviation Rule.

Module 8 | Behavior of Sample Mean (2 of 3)

Reflect and discuss (K)

35. The Federal Pell Grant Program provides need-based grants to low-income undergraduate and certain postbaccalaureate students to promote access to postsecondary education. According to the National Postsecondary Student Aid Study conducted by the U.S. Department of Education in 2008, the average Pell grant award for 2007-2008 was \$2,600. Assume that the standard deviation in Pell grant awards was \$500.

If we randomly sample ________ Pell grant recipients and record the mean Pell grant award for the sample, then repeat the sampling process many, many times, what is the mean and standard deviation of the sample means?

Reflect and discuss (L)

The population mean for Verbal IQ scores is ________, with a standard deviation of 15. Suppose a researcher takes 50 random samples with 30 people in each sample.

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36. What is ____________ of the sample means?

A. 100.
B. 15.
C. 50.
D. 30.
E. Unable to determine from the information provided.

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37. What is the ____________ of the sample means?

A. 15.
B. 2.74.
C. 2.12.
D. Unable to determine.

Module 8 | Behavior of Sample Mean (3 of 3)

Reflect and discuss (M)

38. Recall our earlier scenario: The Federal Pell Grant Program provides need-based grants to low-income undergraduate and certain postbaccalaureate students to promote access to postsecondary education. According to the National Postsecondary Student Aid Study conducted by the U.S. Department of Education in 2008, the average Pell grant award for 2007-2008 was $2,600. Assume that the standard deviation in Pell grants awards was $500.

If we randomly sample ____________ Pell grant recipients, would you be surprised if the mean grant amount for the sample was $2,940? Pick the correct response that gives the best reason.

A. No, $2,940 would not be surprising because this sample result is within 2 standard deviations of the overall mean grant amount of $2,600.
B. Yes, $2,940 would be surprising because this sample result is $340 greater than the overall mean grant amount of $2,600.
C. No, $2,940 would not be surprising because this sample result is only $340 greater than the overall mean grant amount of $2,600, and we expect there to be variability in sample means.
D. Yes, $2,940 would be surprising because this sample result is more than 3 standard deviations from the overall mean grant amount of $2,600.

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39. Which distribution is a plausible representation of 100 samples, with each sample containing ____________ randomly selected students?

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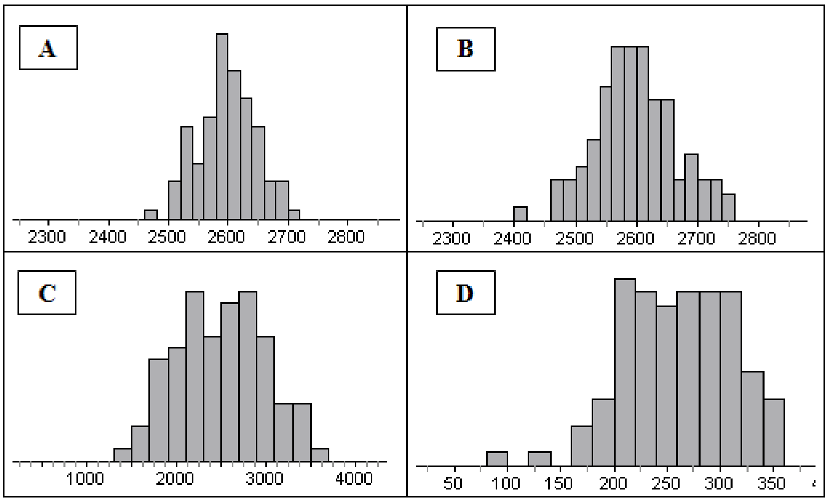
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